SpikeShift.

Traction Outsole for Roundnet

Design Brief

How might we design roundnet-specific cleats that support the sport's rapid, multi-directional movements?

Biomechanical Research

Roundnet Specific Movements



Medial Push-Off



Lateral Landing



Forward Jump

*AI Generated

Pressure Heat Map



Plantar Pressure Data

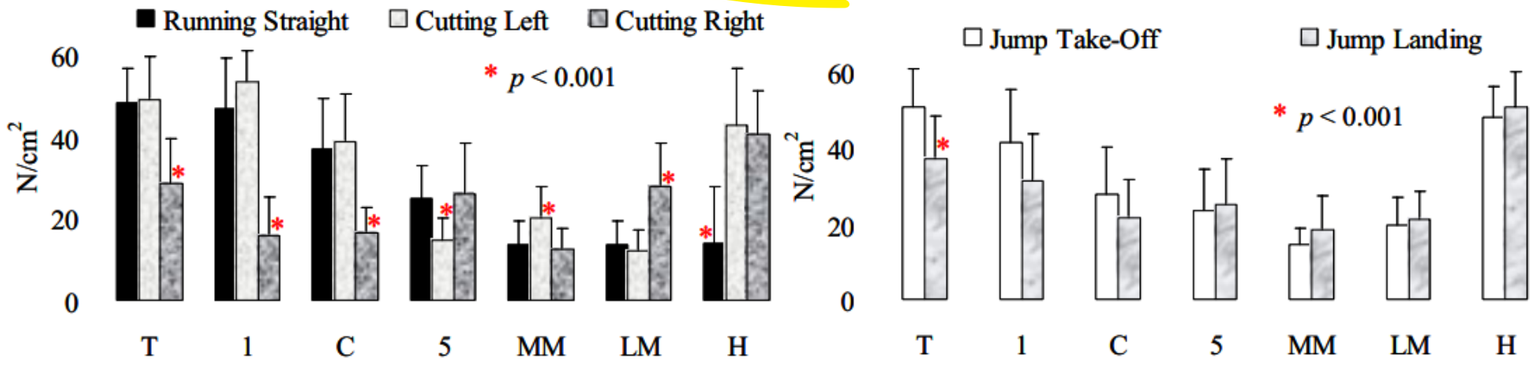
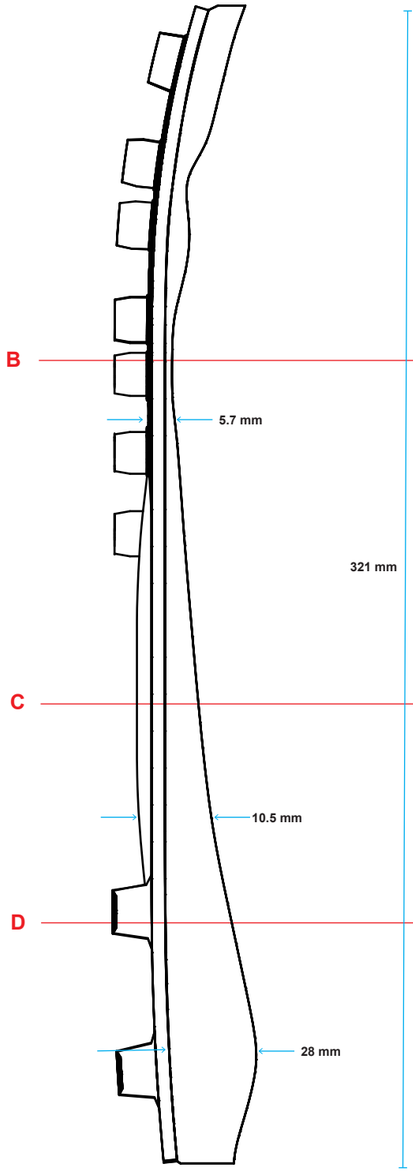


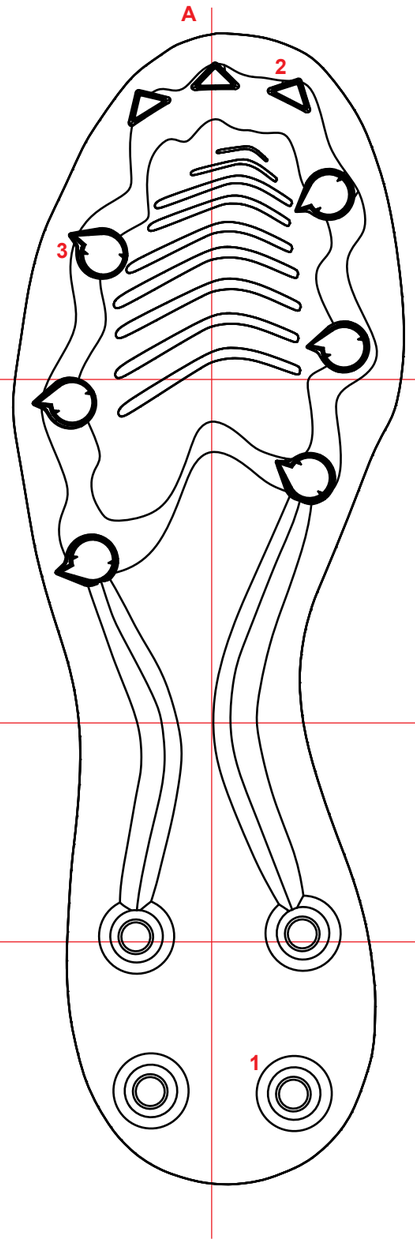
Figure 3: Peak plantar pressure over 7 regions of the right foot during running straight, cutting left, cutting right, jump take-off and jump landing. Significant differences between movements within each region are marked with *.

Orendurff, M., Rohr, E., Segal, A., Kweon, C., Medley, J., Green III, J., & Kadel, N. (University of Washington, Seattle). Regional Foot Pressure During Running, Cutting, Jumping and Landing. Motion Analysis Lab, VA Hospital, Seattle, WA.

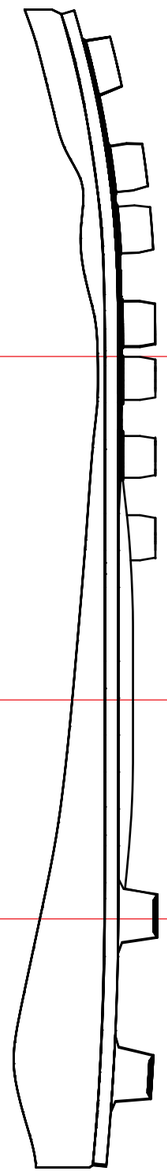
LATERAL VIEW



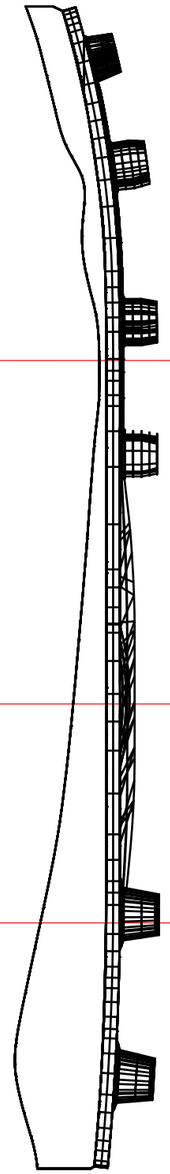
BOTTOM VIEW



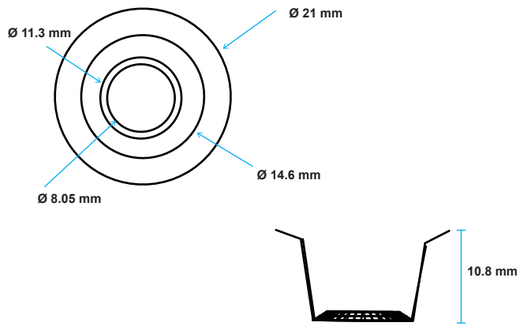
MEDIAL VIEW



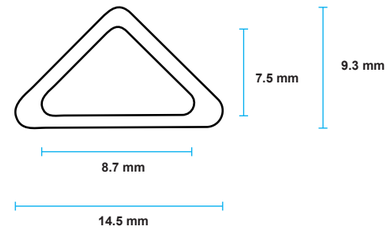
SECTION A-A



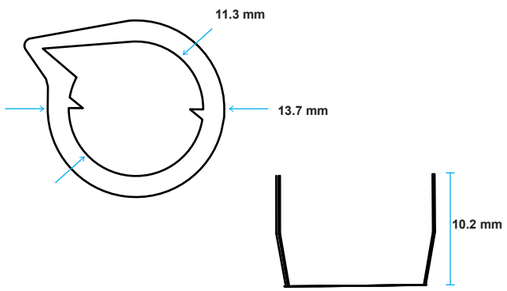
STUD 1



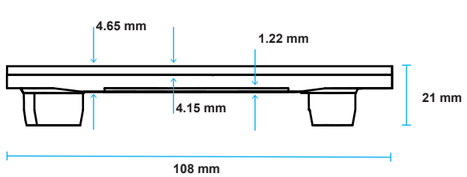
STUD 2



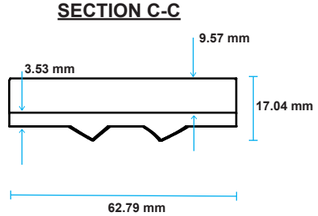
STUD 3



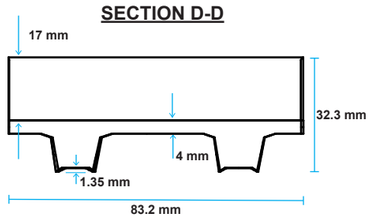
SECTION B-B



SECTION C-C



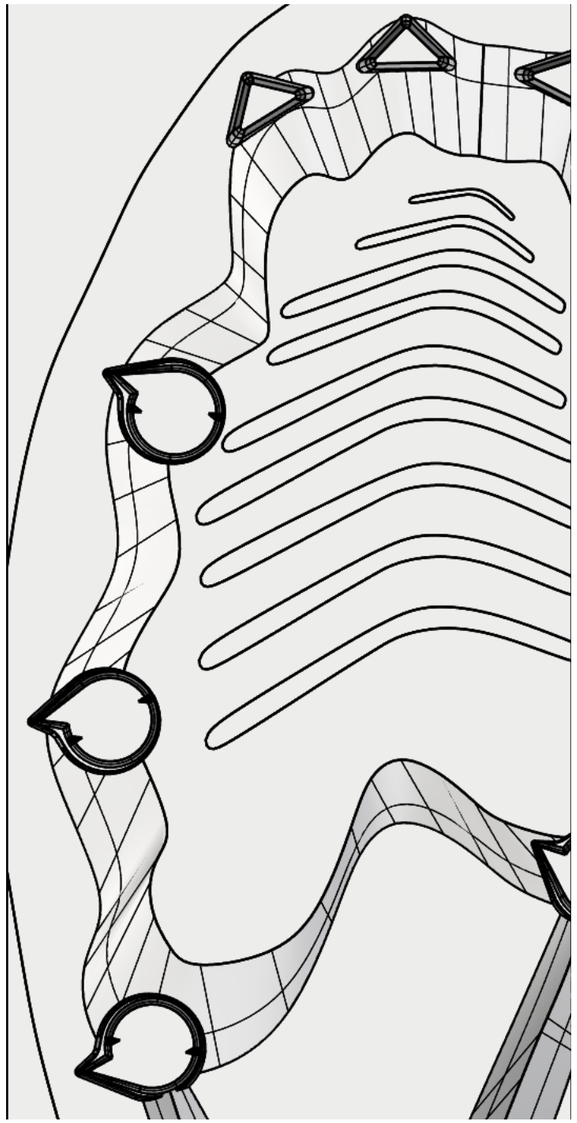
SECTION D-D



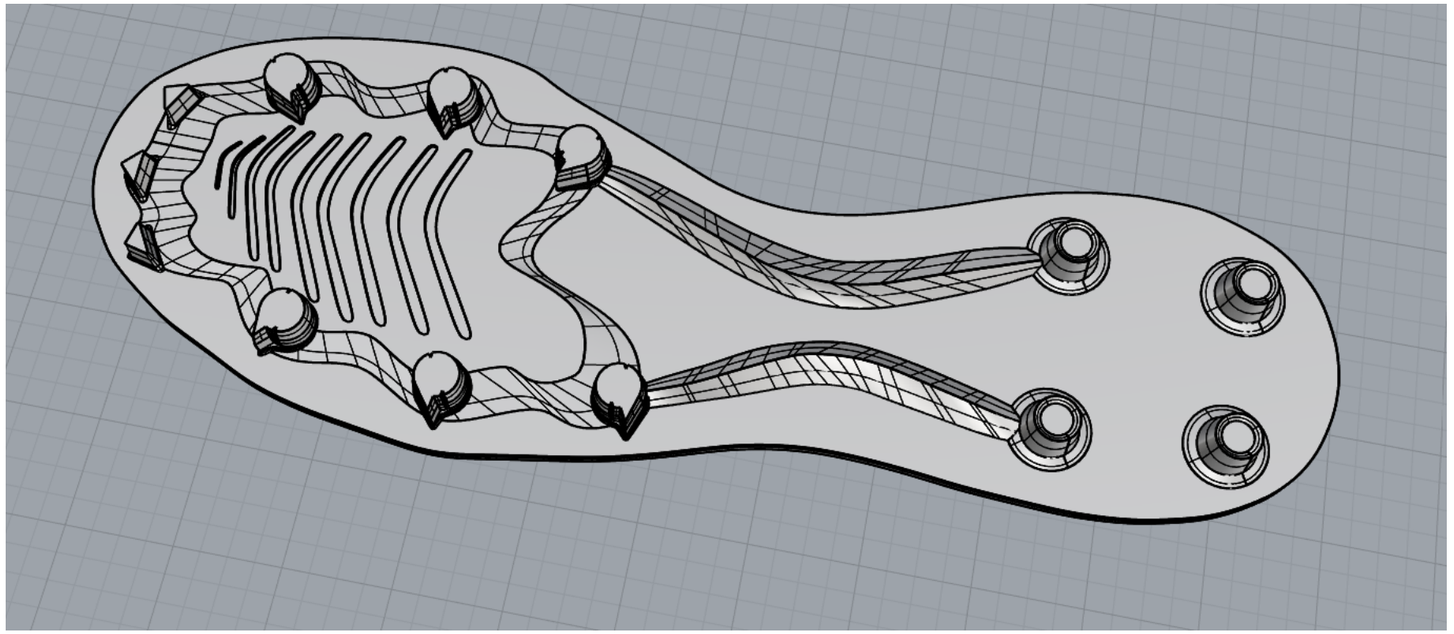
	ELLIOT HILDNER	HW 7	DATE 4/21/26	UNITS mm
	SPORTS PRODUCT DESIGN	ROUNDNET OUTSOLE	SCALE 1:2	PAGE 2

3D CAD

Roundnet Cleat



Testing Slab



Full Model

SpikeShift Render

*via Vizcom



FEATURES & BENEFITS

- Directional medial and lateral studs for roundnet-specific movements
- Bladed studs located on forefoot for explosiveness
- Rounded studs located on heel for rotational stability
- Middfoot support ridges for control
- Flex channels on forefoot for added flexibilty

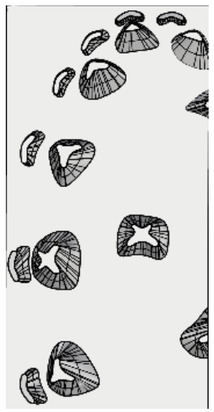
Traction Testing

Baseline vs. SpikeShift

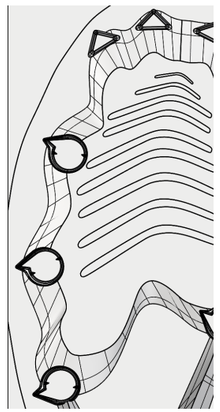
Materials

Traction Slabs - PETG

Baseline Lacrosse



SpikeShift



String



Force Gauge



10lb Dumbbell



Turf Surface



Process

MEASURING

- Lateral Test (Medial & Lateral sides)
- **Force value (N) when traction slab slips**

SET-UP

1. Drill a hole in the center of each traction slab
2. Tie one end of string through hole to slab
3. Tie other end of string to force gauge
4. Place slab studs down on surface
5. Place dumbbell on top of slab
6. Set up video camera for reading values at slipping point

DATA COLLECTION

- Pull force gauge slowly, gradually increasing force
- Record 3 data values per side, for both slabs (12 values in total)



Data

Baseline
SpikeShift

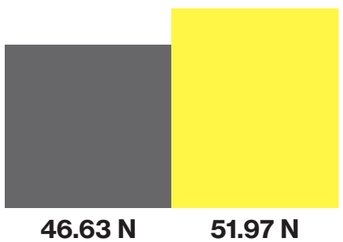
MEDIAL TEST

*averages



LATERAL TEST

*averages



Discussion

- SpikeShift showed higher slip resistance than the lacrosse baseline in both directions, indicating stronger lateral traction.
- Results suggest the stud layout and outsole geometry improve grip for roundnet movements.
- The larger gain in lateral slip resistance suggests stronger grip in that direction.
- While higher traction improves control, excessive grip may limit pivot release and should be balanced.
- Overall, SpikeShift outperforms the baseline, but real-world testing is needed to validate comfort, maneuverability, and consistency.

Design Reflection

What I Learned

- Cleated footwear has a lot of contradictory research claims regarding stud geometry
- Need to design around the idea that you need more traction but also too much traction can cause serious injuries
- Creating a accurate testing method takes longer than the actual testing
- Some brands do not want functional logos
- Testing a section of the outsole doesn’t necessarily give you all of the information you need

References

Orendurff, M. S., Rohr, E. S., Segal, A. D., Medley, J. W., Green, J. R., & Kadel, N. J. (2008). Regional foot pressure during running, cutting, jumping, and landing. The American Journal of Sports Medicine, 36(3), 566–571. https://doi.org/10.1177/0363546507309315